Dataset Description

Data Analysis Midterm Project — Aly Eldeen — 19 July 2026

Dataset Title

Online Retail II

Source

UCI Machine Learning Repository, dataset 502. https://archive.ics.uci.edu/dataset/502/online+retail+ii

Business Domain

E-commerce / Retail.

Number of Rows and Columns

The dataset is published as two yearly files, 2009–2010 and 2010–2011, each with the same 8 original columns. Combined, the raw data totals approximately 1,067,371 transaction rows. After cleaning and feature engineering in Power Query (removing duplicates and adding calculated fields such as Revenue, InvoiceYear, InvoiceMonth, DayOfWeek, IsUK, Transaction Type, CustomerType, and IsOutlier), the working dataset used for this project contains 1,050,856 rows across 18 columns.

Description of Each Variable

Variable	Type	Description	Example
Invoice	Text	Unique 6-digit transaction/order number. Prefixed with 'C' when the transaction is a cancellation.	536365
StockCode	Text	Unique code identifying the product.	85123A
Description	Text	Name of the product.	White Hanging Heart T-Light Holder
Quantity	Numeric (whole number)	Number of units of the product in the transaction. Negative values indicate a return or cancellation.	6
InvoiceDate	Date/Time	Date and time the transaction was generated.	2010-12-01 08:26
Price	Numeric (decimal)	Unit price of the product, in GBP (£).	2.55
Customer ID	Text/Numeric identifier	Unique code identifying the customer. Missing for guest/unidentified transactions.	17850
Country	Text (categorical)	Country where the customer is based.	United Kingdom

During cleaning, additional columns were engineered to support analysis: Revenue (Quantity × Price), InvoiceYear, InvoiceMonth, DayOfWeek, IsUK, Transaction Type (categorizing rows as Successful Sale, Cancelled/Returned, Free Gift/Sample, Inventory Loss/Damaged, or Accounting Adjustment), CustomerType (Identified/Guest), and IsOutlier.

Why This Dataset Was Selected

Online Retail II was selected because it is a large, genuinely messy, real-world transactional dataset. It combines numerical variables (Quantity, Price) with categorical variables (Country, StockCode, Description) in a way that supports meaningful grouping, filtering, and aggregation, and it contains authentic data quality issues — missing Customer IDs, cancelled orders, non-product stock codes, outliers, and inconsistent formatting — that provide a realistic opportunity to demonstrate data cleaning and analytical skills.

Business Problem / Analytical Objective

Which countries, products, and time periods drive the most revenue for this retailer, and how much of that revenue is affected by order cancellations and returns? The analysis uses Revenue, Total Orders, and Average Order Value as measurable business metrics to answer this question and to track performance against defined targets.

Why the Dataset Satisfies the Project Requirements

Requirement	How it is satisfied
Structured tabular data	Rows and columns, no restructuring needed
Real-world domain	E-commerce / Retail
≥ 1,000 records	≈ 1.05 million rows after cleaning
≥ 8 variables	8 original columns, 18 after feature engineering
Numerical and categorical variables	Quantity/Price/Revenue are numerical; Country/StockCode/Description are categorical
Measurable business metric	Revenue, Total Orders, and Average Order Value
Real-world imperfections	Missing Customer IDs, cancellations, non-product codes, outliers, inconsistent text, wrong data types
Supports EDA, dashboards, and business insight	Demonstrated in the accompanying project report and Power BI dashboard